Spot a Keyword

For the next seven chapters you build ELIZA, the first chatbot. Before it can answer, it has to notice the word in your sentence that it knows something about.

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Plan: seven chapters, one chatbot

Chapter	ELIZA learns to ...
7 Iteration and search	**spot a keyword, give a first reply**
8 Strings, regular expressions	turn "I am X" into "Why are you X?"
9 Lists	hold many rules and replies, remember
10 Dictionaries	swap "my" for "your", count what it does
11 Tuples	rank its rules, print a transcript
12 Text analysis and generation	find its blind spots, generate replies
13 Files and databases	keep its script and memory on disk

ELIZA, 1966

- You talked to ELIZA in week 2. Its best-known script, DOCTOR, imitates a therapist who turns your words back into questions
- Written by Joseph Weizenbaum at MIT, 1964–1966
- The program was written in **MAD-SLIP**, a list-processing extension of the language MAD, on an IBM 7094
- The scripts were written as nested lists that look like Lisp – the program itself was not Lisp
- The original source code was found in Weizenbaum's papers at MIT in 2021

How ELIZA answers

1. Look for a **keyword** in the sentence
2. Pick the rule that belongs to the keyword
3. Build the reply from the rule (and from pieces of your sentence)
4. No keyword? Say something neutral: "Please go on."

You type	Keyword	ELIZA answers
My mother is always busy.	mother	Tell me more about your family.
I had a strange dream.	dream	What does that dream suggest?
The weather is nice.	(none)	Please go on.

Try these sentences: `masswerk.at/elizabot`

Today is step 1: find the keyword. That takes loops.

Loop over a string

→ **Switch to editor · code-along 1**

```
for letter in "Why?":
    print(letter, end=" ")
```

```
W h y ?
```

The **loop variable** `letter` holds one character per pass. `end=" "` replaces the line break with a space.

Four answers. ELIZA needs **one**: is it a question or not?

How could you do that?

Is the sentence a question? Check every letter:

```
for letter in "Why?":
    if letter == "?":
        print(True)
    else:
        print(False)
```

```
False
False
False
True
```

One answer: print becomes return

→ **Switch to editor · TRY (2 min) · predict, then run**

```
def has_question_mark(sentence):
    for letter in sentence:
        if letter == "?":
            return True
        else:
            return False

print(has_question_mark("Why?"))
```

```
False
```

How would you fix it?

Scaffold `print` as the first line of the loop body:

```
    for letter in sentence:
        print("checking", letter)
```

```
checking W
False
```

→ The loop checks `"W"` and nothing else. The `else` runs `return False` – and `return` ends the whole function, and the loop with it.

The fix: "not found" goes after the loop

```
def has_question_mark(sentence):
    for letter in sentence:
        if letter == "?":
            return True
    return False

print(has_question_mark("Why?"))
print(has_question_mark("I am tired."))
```

```
True
False
```

- One `"?"` is enough: `return True` stays in the loop
- "No question mark" is only true after you have looked at *every* letter: `return False` goes after the loop
- `print` does not end anything; `return` does
- Question: what if you indent `return False` under the `if` instead of the `for`? What do you get, and why?

Count with a loop

→ **Switch to editor · code-along 2**

One more word than there are spaces:

```
def count_words(sentence):
    count = 1
    for letter in sentence:
        if letter == " ":
            count = count + 1
    return count

print(count_words("My mother is always busy."))
```

```
5
```

- count is a **counter**: **initialized** before the loop, updated inside it
- Short form: `count += 1` (**augmented assignment**), also `-=`, `*=`
- TRY (2 min): remove `count = 1`. What happens?

```
UnboundLocalError: local variable
'count' referenced before assignment
```

→ To update count, Python needs its old value – there is none.

Question: where does count_words go wrong?

What do these calls return? (The first one has two spaces.)

```
print(count_words("Hello  there"))
print(count_words(""))
```

```
3
1
```

→ Both wrong: the function counts spaces, not words. In chapter 9, `split` does it properly.

Is the word in the sentence?

→ **Switch to editor · code-along 3**

```
print("mother" in "My mother is always busy.")
print("father" in "My mother is always busy.")
print("mother" in "Mother is always busy.")
```

```
True
False
False
```

With strings, `in` tests for a **substring**. It is case-sensitive: `"Mother"` is not `"mother"`.

Lower the sentence first:

```
def mentions(sentence, word):
    return word in sentence.lower()

print(mentions("Mother is always busy.",
               "mother"))
print(mentions("Mother is always busy.",
               "Mother"))
print(mentions("Mother is always busy.",
               "MOTHER".lower()))
```

```
True
False
True
```

Why `False` in the middle?

Lower both sides

- `mentions` lowers only the sentence. `"Mother"` keeps its capital M and can never match
- The third call works because the caller lowers the word – but then every caller has to remember it
- Let the function do the work: lower both sides

```
def mentions(sentence, word):
    return word.lower() in sentence.lower()

print(mentions("Mother is always busy.", "mother"))
print(mentions("Mother is always busy.", "Mother"))
print(mentions("Mother is always busy.", "MOTHER"))
```

```
True
True
True
```

`lower()` is a **method**: a function called on a value. The comparison is already `True` or `False` – return it directly.

in finds letters, not words

→ **Switch to editor · TRY (2 min) · predict, then run**

```
print(mentions("The pillow smothered the sound.", "mother"))
print(mentions("I know.", "no"))
print(mentions("I like computers.", "computer"))
```

```
True
True
True
```

- "mother" is inside "smothered", "no" inside "know" – wrong
- "computer" inside "computers" – here it helps
- ELIZA lives with this until chapter 8: regular expressions can match whole words

Keywords belong in a file

→ **Switch to editor · code-along 4 · setup cell, run once**

```
open("keywords.txt", "w").write("""sorry
remember
dream
computer
mother
father
sister
brother
always
because
""")
```

You learn later how this works.

```
file = open("keywords.txt")
print(repr(file.readline()))
print(repr(file.readline()))
```

```
'sorry\n'
'remember\n'
```

- open returns a **file object**; `readline` reads one line
- Every line ends with a **newline** `\n`; `repr` shows it
- The file object remembers where it stopped

Read every line: `for` and `strip`

```
for line in open("keywords.txt"):
    print(line)
```

```
sorry

remember

dream
...
```

`\n` at the end of the line, plus the line break of `print`

```
for line in open("keywords.txt"):
    word = line.strip()
    print(word, end=" ")
```

```
sorry remember dream computer mother
father sister brother always because
```

`strip()` removes whitespace – spaces, `\n` – at both ends

Can you put the pieces together?

For each keyword in the file, check whether it is in the sentence:
`show_keywords(sentence)`

Which keywords are in the sentence?

```
def show_keywords(sentence):
    for line in open("keywords.txt"):
        word = line.strip()
        if word in sentence.lower():
            print(word)

show_keywords("My mother says I always "
              "dream about computers.")
```

```
dream
computer
mother
always
```

The keywords come out in the order of the **file**, not of the sentence.

Which one should ELIZA answer? That is the next question.

Practice: how many keywords?

→ **Switch to editor · 10 min, then 5 min discussion**

Write `count_keywords(sentence)`: it *returns* how many keywords from `keywords.txt` appear in `sentence` (ignore case). Use a counter.

```
>>> print(count_keywords("My mother and my father never talk."))
2
>>> print(count_keywords("Sorry, I forgot."))
1
>>> print(count_keywords("The weather is nice."))
0
```

Checkout

1. Why must `return False` in `has_question_mark` come after the loop?

→ "Not found" is only true after every character was checked. Inside the loop, the `else` returns after the first one.

2. Why does `print(line)` in a loop over a file print empty lines?

→ Each line ends with `\n`, and `print` adds another line break. Use `line.strip()`.

Summary

- A `for` loop over a string visits one character per pass; the **loop variable** holds the current one
- A `return` inside a loop ends the function at once. "Not found" goes after the loop
- A **counter** is initialized before the loop and updated inside it; `count += 1` is short for `count = count + 1`
- `in` tests whether a string is a **substring** of another. It is case-sensitive – use `lower()` – and it also finds a word inside another word
- `open` returns a **file object**; `readline` reads one line, a `for` loop reads all of them. Each line ends with `\n`; `strip()` removes it

Glossary: terms from session 1

#	Term	Meaning
1	chatbot	Program that converses in natural language
2	keyword	Word that tells ELIZA which rule to use
3	loop variable	Variable that takes a new value in each pass
4	counter	Variable that counts, e.g. `count += 1`
5	initialize	Give a variable its first value
6	augmented assignment	Update in place, e.g. `+=`
7	substring	String that is part of another string
8	method	Function called on a value, e.g. `s.lower()`
9	file object	What `open` returns; reads a file line by line
10	newline	Character `\n` that ends a line

Next: the first reply

show_keywords prints every keyword it finds. ELIZA needs exactly one.

- A **search** that stops at the first keyword
- ELIZA's first replies – and a fallback for everything else
- Tests written into the docstring